Indian Institute of Science Education and Research, Kolkata
Math Methods (PH2204)
Class Test -II, Date: Feb 22, 2025

Timing: 8:00 to 09:15 | Spring 2025 | Full Mark: 100

Attempt all *five* questions. Each carris 20 marks.

Notation: $z = x + iy = re^{i\theta}$, where $x = r\cos\theta, y = r\sin\theta$ and $i = \sqrt{-1}$. $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \equiv (x, y, z)$. $\nabla \equiv (\partial_x, \partial_y, \partial_z)$. Spherical polar coordinates: $\vec{r} = (r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta)$; $\nabla \equiv \left(\partial_r, \frac{1}{r}\partial_\theta, \frac{1}{r\sin\theta}\partial_\phi\right)$.

1. **(a)** Express Cauchy-Riemann condition in polar coordinates. **[5]**
   (b) Find if $f(z) = zz^* \exp(\frac{1}{z^*} - \frac{1}{z})$ is an analytic function. **[7]**
   (c) In what region of the complex plane $f(z) = |x| - i|y|$ is analytic? **[8]**

2. **(a)** Find the real and imaginary part of i^i and $\log(-1)$. **[6]**
   (b) If $z = e^{i\theta}$, then show $z^n + \frac{1}{z^n} = 2\cos(n\theta)$. Use this to show $4\cos^3(\theta) = \cos(3\theta) + 3\cos(\theta)$. **[6+8]**

3. **(a)** Find the Taylor series of $f(z) = \sum_{k=1}^{\infty}(-1)^{k+1}\sin(\frac{z}{k})$ about $z = 0$. What is its radius of convergence? **[8+4]**
   (b) What type of singularities (if any) possessed by (i) $f(z) = \frac{1}{z-3}$, (ii) $f(z) = \sinh(1/z)$, at $z = 0$ and $z = \infty$. **[4+4]**

4. Find α so that: **(a)** $\nabla\left(\frac{1}{r}\right) = -r^\alpha\vec{r}$, **(b)** $\nabla\cdot(r\vec{r}) = \alpha r$, **(c)** $\nabla\cdot(\nabla(r^3)) = \alpha r$, **(d)** $\nabla\times\nabla(r^\alpha) = \nabla\cdot\nabla\times(\sin(\alpha r)\vec{r})$ **[20]**

5. Use Levi-Civita symbols to prove: **(a)** $\nabla\cdot(\vec{A}\times\vec{B}) = \vec{B}\cdot(\nabla\times\vec{A}) - \vec{A}\cdot(\nabla\times\vec{B})$ **(b)** $n\times n$ square matrices A, B, C, D satisfy $\det(ABCD) = \det(DABC)$. **[10+10]**

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